

Evaluating Income-Related Performance Gaps in Vision-Language Models

Computer Vision Course Project | Research Proposal | Domain: Multimodal AI Systems

Chosen Domain

Multimodal AI Systems, focusing on pretrained vision-language models for zero-shot household-object classification and image captioning.

Problem Statement

Everyday objects can look very different across homes because of materials, design, and living conditions. A model familiar with one visual form of a stove, roof, or light source may fail on another. This project tests that robustness problem under a controlled design: CLIP performs six-way zero-shot classification and BLIP generates captions for the same balanced set of household images. Performance will be compared across income quartiles without assuming in advance that a gap exists.

Motivation

- Uneven recognition can affect image search, accessibility tools, and other applications that rely on general-purpose visual models.
- Balancing every object category across income groups separates the main comparison from simple class-frequency differences.
- Pretrained models and a fixed small subset make the study feasible while still supporting quantitative and qualitative analysis.

Research Question

Primary: For six category-balanced household-object concepts, how do CLIP top-1 accuracy and BLIP object-caption correctness vary from the lowest (Q1) to highest (Q4) income quartile? **Secondary:** Does the label-free prefix "the main household object in this image is" change BLIP's overall and Q1 correctness without increasing hallucination?

Dataset or Data Source

The study uses the 1,600-row `nlphuji/dollar_street_test` split derived from Dollar Street [3], which provides image, country, region, object, and monthly household-income metadata. The fixed subset contains 168 unique images from 44 countries: roof, light source, stove, trash container, switch, and footwear, with seven images in each category x quartile cell (6 x 4 x 7). Ambiguous multi-class rows are excluded. Broader study labels and predeclared synonyms avoid penalizing culturally valid forms such as lanterns or sandals.

Planned Method

- **Data control:** validate IDs, files, labels, and numeric income; compute quartiles once; retain the fixed balanced manifest for every run.
- **CLIP baseline [1]:** use `openai/clip-vit-base-patch32` with six prompts of the form "a photo of a {label}"; no training or prompt tuning.
- **BLIP comparison [2]:** use `Salesforce/blip-image-captioning-base` with identical deterministic decoding for an unprompted caption and the single label-free prefix.
- **Metrics:** report category-macro top-1 accuracy and correct-label rank for CLIP; for BLIP, report accepted-term recall plus human correctness and hallucination rates.
- **Human review:** both team members will independently rate all 336 captions while blinded to income quartile and prompt condition. Cohen's kappa and adjudicated disagreements will be reported.

- **Analysis:** compare Q4-Q1 and maximum-minimum gaps with 95% stratified bootstrap confidence intervals. Prompt effects use paired bootstrap intervals. Regional results are descriptive only and always include sample counts.

Expected Outcome

The project will produce a reproducible, category-controlled account of where CLIP and BLIP succeed or fail across income groups. The result may show a gap, no stable gap, or category-specific variation; the prompt may improve, leave unchanged, or worsen captions. The contribution is the measured comparison, uncertainty, prompt-effect estimate, and failure taxonomy - not a predetermined fairness claim.

Planned Interactive Demo Idea

A hosted Gradio webpage (usable without local installation) will let users select a curated image or upload one, inspect CLIP label scores, and compare unprompted and prompted BLIP captions side by side. Aggregate charts will filter by category, income quartile, and region and will show sample counts, confidence intervals, and representative failures. User uploads without verified income metadata will be shown as examples but excluded from subgroup statistics.

Feasibility and Scope

No model training is required. The 168 images, manifest, evaluation code, and a four-image end-to-end smoke test are already complete. The full run, double review, hosted demo, and analysis are achievable by 14 September. Conclusions will be limited to the selected models, categories, and test split; income associations will not be presented as causal effects.

Reproducibility and Validity Controls

Image IDs, selection rules, model checkpoints, software versions, prompt strings, decoding settings, and random seeds will be fixed and logged before the full evaluation. Labels and accepted synonyms will not be changed after model outputs are inspected. Every CLIP score, caption, automated judgment, and adjudicated human label will be saved in analysis-ready CSV files. Before rating begins, the team will agree on written definitions of a correct object reference, a non-informative caption, and a hallucination. Independent ratings will be locked before disagreements are discussed.

Risks and Responsible Reporting

Household income is contextual metadata, not evidence that income causes a prediction. The small, deliberately balanced subset cannot represent every culture, object type, or vision-language model, and category selection may introduce selection bias. We will therefore avoid country rankings, keep regional comparisons exploratory, and report denominators beside every subgroup result. Accepted-term recall can miss valid wording or reward a keyword in an otherwise incorrect caption, so conclusions will be checked against blinded human ratings. Dollar Street's CC-BY 4.0 attribution will be retained; the repository will provide access instructions instead of redistributing the full dataset, and demo examples will include source attribution.

References: [1] Radford et al., ICML 2021 (CLIP). [2] Li et al., ICML 2022 (BLIP). [3] Gaviria Rojas et al., NeurIPS 2022 (Dollar Street).